PeerAhammad M Bagawan

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GitHub | LinkedIn | ORCID | Portfolio

Profile

Computer Science undergraduate and ML researcher with a NeurIPS 2026 accepted poster and an arXiv preprint. Experienced in model development, experimental design, statistical evaluation, and AI application engineering. Research interests include sequential learning, state-space models, and capability dynamics.

Education

Visvesvaraya Technological University (VTU) **2023 – Expected 2027**
B.Tech. in Computer Science and Engineering *Belagavi, Karnataka*
CGPA: 7.2/10 (through Semester 6)

Publications

- **When Does Sequential Detection Collapse to a Scalar? A Necessary and Sufficient Characterisation.** P. S. Hiremath, **P. M. Bagawan**, S. Bhekane. *NeurIPS 2026 (Poster, accepted).* [OpenReview] *Role: implementation and experiments.*
- **PARD-SSM: Probabilistic Cyber-Attack Regime Detection via Variational Switching State-Space Models.** P. S. Hiremath, **P. M. Bagawan**, S. Bhekane. *arXiv preprint.* [arXiv] *Role: model implementation and experiments.*
- **The Measure Gap: Information-Theoretic Limits of Deterministic Belief Encoders in POMDPs.** P. S. Hiremath, **P. M. Bagawan**, S. Bhekane. *Manuscript in preparation.*

Research Experience

Research Intern, Capability Dynamics **Sep 2026 – Present**
VTU, Belagavi *Supervisor: M. N. Birje*

- Study learning dynamics and emergent capabilities, focusing on the distinction between capability acquisition and observable expression.
- Conducted a 30-seed grokking pilot using one-block Transformers on modular addition (mod 97); investigated initial spectral competition as a potential predictor.
- Identified commutativity leakage at the first 95%-training-accuracy checkpoint (91.30% vs. 0.316% subgroup accuracy) and designed a corrected, symmetry-aware evaluation protocol.

Selected Projects

PARD-SSM: Probabilistic Attack Regime Detection GitHub
Python, PyTorch, Switching State-Space Models, scikit-learn

- Reproduced and extended a hybrid Switching SSM–Random Forest detector on CICIDS2017 and UNSW-NB15.
- Recorded F1 scores of 96.07% (CICIDS2017) and 94.38% (UNSW-NB15) on 50,000-sample evaluation subsets; conducted baseline comparisons and ablation studies.

RAPO: Regret-Aware Policy Optimization GitHub
Python, PyTorch, Reinforcement Learning

- Developed regret-aware policy optimization experiments across graph diffusion, grid-world navigation, and recommendation tasks.
- Analysed replay-induced harm and compared policy behaviour against stationary baselines.

ThreatFlow: Streaming Cyber-Threat Detection GitHub | Live Demo
Python, FastAPI, WebSockets, scikit-learn

- Built a streaming detection pipeline combining Switching SSM context with Random Forest classification.
- Achieved 96.02% F1 and 99.823% ROC-AUC on a 50,000-flow chronological evaluation.

TraceMind AI: AI Agent Trace Analysis Platform GitHub
Python, FastAPI, React, TypeScript

- Built a platform for analysing AI-agent execution traces and benchmarking agent runs.
- Implemented trace and benchmark API workflows, backed by a passing suite of 63 backend tests.

Technical Skills

Languages	Python, JavaScript
ML	PyTorch, JAX, NumPy, Pandas, scikit-learn
Development	FastAPI, WebSockets, React, TypeScript, Git
Research	Experimental Design, Statistical Evaluation, Sequential Detection, State-Space Models